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The Ghost and the Machine

Jonathan MARTIN & Anthony CHEMERO*

As Rockwell points out in “Representation and Radical Empiricism,” this issue, theoretical progress (or, at least, clarification) in philosophy is often the happy result of the kind of friendly adversarial relationships that characterize academic philosophical debates. Rockwell's paper is a case in point; he develops a theory of direct perception that bears some similarities with those defended in Chemero's *Radical Embodied Cognitive Science* (RECS) and even comes to some similar conclusions, but it is precisely where he misunderstands Chemero that we find an opportunity to clarify what it is that RECS is all about. Rockwell presents his view as a slightly tempered endorsement of RECS that agrees that “minds and worlds are blended in a constantly shifting dynamic relationship, and that perception does not function by creating representations inside the brain,” but he expresses misgivings about some of Chemero's more radical claims and offers some correctives (Rockwell, p. 220). The problem is that both his misgivings and his suggested correctives betray a misunderstanding of some of the key features that make RECS a departure from traditional cognitive science – most importantly, the claim that nonmechanistic, dynamical explanations of perception, action, and cognition are genuine explanations. Among the many things we share with Rockwell is a fondness for Gilbert Ryle. When Rockwell named his book *Neither Brain Nor Ghost*, he was referring to Ryle's famous “ghost in the machine”. Of course, we agree with Rockwell that cognitive science should not in the business of studying ghosts. What Rockwell fails to realize, though, is that according to RECS, cognitive science should not therefore be condemned to studying machines. Before explaining what this means, we will look at a few of the more important issues where it seems that Rockwell and Chemero disagree.

SOME BRIEF (BUT IMPORTANT) CLARIFICATIONS

Rockwell writes that, while he agrees that cognition involves a fluid interaction between animal and environment and that perception does not (necessarily) involve mental representation, he has “always had trouble with Chemero's claim that these facts permit us to deny the existence of representations altogether, which claim is sometimes expressed by saying it is possible to do cognition without what Chemero calls 'mental gymnastics' (*ibid.*, p. 220).” There are two major claims to separate out here: 1) The dynamic interaction between agent and environment permit us to deny the existence of representations and 2) It is possible to do cognition without mental gymnastics. We bring attention to claim 1 because it is a claim that Chemero (2009) goes to some lengths *not to endorse*; chapter 1 is directed entirely at the problems associated with philosophical arguments that purport to demonstrate the futility

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of certain scientific research programs/methodologies. And, as chapter 3 of the book argues, given the myriad uses to which the concept of “representation” has been put in cognitive science, we lack any clear notion of representation to be eliminativist *about*. Given that so many fundamentally different sorts of things have been called representations in the history of cognitive modeling, it's not clear what kind of thing's existence the eliminativist would be denying. Simply put, to read Chemero (2009) as full-fledged eliminativism about representation is to miss the point; the primary purpose of the book is to explore the numerous areas of interest to cognitive scientists that are being explored without the use of mental gymnastics, to defend the explanatory power of these non-traditional models, and to see what light is shed on various philosophical issues by these approaches.

On the other hand, claim 2) is ambiguous – it depends on how we want to read “possible to do cognition.” On one way of reading this phrase, claim 2) is a claim about the very *possibility* of cognitive or perceptual processes that do not in any way involve “mental gymnastics.” The second way of reading it is that it is a claim about the viability of a certain research methodology; specifically, that there exist behaviors that are correctly classified as cognitive that can be given a satisfactory explanation without reference to “mental gymnastics.” While both of these readings seem consistent with RECS, Chemero makes it clear that his emphasis lies on examining models that do not implicate mental gymnastics in explanations – a fact that favors the second reading. In short, one of the main (and most “radical”) commitments of RECS is that there are adequate *explanations* and *models* of cognition, action and perception that do not implicate mental gymnastics, not that there is no place for mental gymnastics in cognitive science. And, while he does spend some time arguing that the explanatory utility and scope of traditional representational accounts' use of the term “representation” has been overestimated, Chemero (2009) is primarily an account of cases where ecological and dynamical explanations take the place of mental gymnastics; it's “Radical embodied cognitive science,” not “Radical embodied cognition” for a reason. The legitimacy of this replacement will occupy a large part of the later part of this response. But, before we get to the issue of cognition versus cognitive science, we will spend some time attending to why Rockwell seems to have trouble with this: if he is going to make suggestions for repairing some defect of Chemero's picture, the alleged defects have to be features of a view that is recognizable as the one defended in *RECS*.

MENTAL GYMNASTICS AND MAGIC

With these clarifications out in the open, what is it exactly that Rockwell takes issue with here? His main difficulty seems to be that he thinks that the above characterization of RECS is problematic because it lacks a clear definition of “mental gymnastics.” “Without a clear definition,” he argues, “any physical activity that accounts for cognition could arguably fit the bill, which turns anti-representationalism into a *de facto* defense of magic (p. 220).” This is a pretty weighty charge if it is right (and if we, again, ignore the claim that Chemero endorses anti-representationalism), and this is one of the main areas towards which Rockwell aims his intended correction. It is also, as we'll

see, where he begins to decisively (albeit unwittingly) part ways with the central commitment of RECS. As he points out, Chemero *does* at various points tell us what he means by “mental gymnastics”: “The construction, manipulation and use of representations of the world (Chemero, 2009, p. 8).” But, he argues, this definition shouldn’t satisfy us because it relies on the notion of “representation” that, as both Rockwell and Chemero seem to agree, can be used to mean almost anything.

So, whence comes the allegation that the radical embodied cognitive scientist must believe in magic? Here’s the gist of it, as we understand it: Lacking a precise definition of what we mean by “mental gymnastics,” the radical embodied cognitive scientists who claims to obviate the need for mental gymnastics-involving explanations of cognition, action, and perception will be forced to embrace some kind of magic in their explanations, as any physical process (in the brain, presumably) that explains the cognitive phenomenon in question could be seen as an instance of mental gymnastics. Therefore, the radical embodied cognitive scientist either must deny that their dynamical models provide explanations or they must hold that the phenomena are explained by magic, as no physical process has been tasked with the job of *explaining* those dynamics. Regarding this problem, Rockwell suggests that Chemero is being deliberately evasive and mysterious; when Lee & Reddish explain the diving gannet’s perception of time-to-contact with the water in terms of the dynamic interplay of the bird’s optic flow field, the bird’s movement, and information available in the ambient light (*i.e.* not *computing* time-to-contact; Lee & Reddish, 1981), they have *left out* the neural machinery that *really explains* this ability. If we don’t implicate *some* neural machinery in our explanation, that is, the ability for the bird to perceive time-to-contact is just a mystery and Chemero’s insistence that in certain instances “one can simply *see...* how fast something is moving without computing it (Chemero, p. 122)” as Rockwell puts it, “seem[s] to be aggressively mysterious, with the italics functioning as a kind of rhetorical table-pounding (Rockwell, p. 220).”

IT’S NOT A BUG...

In response to Rockwell’s diagnosis of a major problem in RECS in need of fixing, we offer our own diagnosis: what Rockwell means by “explanation” just is mechanistic explanation. You can, by Rockwell’s lights, eliminate the ghost, but you need to explain the machine. This is the deep disagreement between Rockwell and advocates of RECS. What Rockwell sees as a bit of evasive trickery is really just Chemero’s embrace of one of the core commitments of RECS – the commitment to dynamical explanations as genuine, nonmechanistic explanations. Looking at Rockwell’s attempt to fix this perceived gap in RECS supports the conclusion that what he’s looking for are mechanistic explanations. He suggests that what Chemero *ought* to mean by “mental gymnastics” are the linguiform, symbolic representations that populate computational theories like Fodor and Pylyshyn’s Representational Theory of Mind (RTM), classical AI, and other approaches that treat cognition as the algorithmic manipulation of concatenative, recombinatorial, and context-invariant symbols in a serial processing architecture. He then introduces a distinction between these types of representations (he calls them

“computational”) and analog representations – representations that represent in a way analogous not to sentences but to maps or pictures. For Rockwell, this distinction is significant because the kinds of representations he suggests we call “mental gymnastics” are those performing formal inferences on perceptual information, whereas the analog representations formed during the successful training of artificial neural networks (ANNs) are not fruitfully understood as involving the kind of discrete inferential stages associated with computational information processing.

Understanding the work that Rockwell thinks this does in correcting RECS's allegedly promiscuous understanding of mental gymnastics requires some explication of the picture he wishes to supplement it with (we will assume some basic familiarity with 1990s-style connectionism for brevity's sake). The analog representations that carve out the conceptual terrain of ANNs represent by partitioning the phase space of possible hidden-unit activations in a way that maps every possible pattern of activation at the network's input layer with an appropriate pattern of activation at the output layer (as defined by the training set and the learning algorithm employed). This mapping between inputs and outputs in successfully trained ANNs enables these networks to navigate the cognitive task it was trained to perform. In a well-trained ANN the configuration of connection weightings between its layers of units enables it to perform well on novel inputs (inputs not found in the training set), indicating that the network has developed some *know-how* of its task, rather than just performing rote associations between its training set and the outputs designated “correct” during training. This ability to generalize is where connectionism's analog representations are thought to pull their explanatory weight; the partitioning of the hidden-units' phase space form attractors that reflect context-sensitive (i.e. relative to the task and training history) statistical regularities across diverse inputs, thus enabling it to group quite different inputs as *of the same kind* in some task-relevant respect. These representations, distributed across every connection in the network, are what Garrison Cottrell and Janet Metcalfe call “holons” when they encode useful information about the ANN's entire receptive field (input layer) in a way that defies assigning to any specific units the job of detecting a particular feature (Cottrell & Metcalfe, 1991, p. 570).

For Rockwell, holon representations provide a model of perception that is consistent with a form of direct perception; holons map perceptual inputs to behavior in a way that is non-inferential (and thus direct) but that explains how a trained ANN can perceive salient features of its environment in a way that connects the perception of certain features directly to skilled behavior. Because the partitioning of a network's conceptual space in the way described above is thought to explain the ANN's acquisition of task-appropriate perceptual representations while tying representation inextricably to action, Rockwell sees these analog representations as a model for explaining how organisms might directly perceive affordances. He writes that ANNs are “actually truncated affordance systems” and that “[knowing-how] to make perceptual judgments about objects sitting in front of you is just one more skill performed by vector transformations of perceptual space into behavioral space (Rockwell, pp. 225-226).” This analysis is meant to serve two main purposes. First, it supplies a

serviceable definition for “mental gymnastics” (i.e. symbolic computation) that does not exclude processes that are consistent with RECS's commitment to dynamical/Gibsonian direct perception. And, second, it is meant to put such processes to work in explaining the ability of such direct perception to govern situated action.

But, as we hinted earlier, the solution is not really a solution at all; it is a misunderstanding. Rockwell's insistence on augmenting RECS's conceptual repertoire with analog representations such as connectionist hidden-unit weightings, holons, and the like reveals a commitment to explaining cognition, action, and perception in terms of neural machinery (assuming, as he does, that first generation ANNs shed biologically plausible light on neural mechanisms). And, while the representational scheme developed around parallel distributed processing does differ in important respects from classical symbolic computationalism, it still is committed to explaining cognitive capacities in terms of mechanisms, whether those mechanisms are comprised of idealized connectionist nodes or real neural populations. And, there's the rub: Rockwell thinks that RECS requires magic because he thinks that dynamical and ecological explanations of cognition and perception are themselves *legitimated* only by being given a mechanistic interpretation. In other words, his solution to a supposed problem for RECS is to abandon one of the defining features of RECS: the claim that nonmechanistic dynamical explanations are genuine explanations. You can debate whether chess or Trivial Pursuit are more rewarding diversions, but it makes no sense to argue that Trivial Pursuit is deficient due to a lack of emphasis on king safety. For the advocate of RECS, this commitment is non-negotiable, though it is controversial, and the rest of our paper will be concerned with defending it.

To be fair, there is a lot of Rockwell's account that we have not addressed. In the course of presenting his correctives he sketches a bold (empirical) neuroscientific theory of direct perception that draws on work as diverse as Wittgenstein, Dewey, along with some dubious pop-neuroscience about brain lateralization. (Although Rockwell assures us that his pop neuroscience on brain lateralization is so well established as to not require specific citation, the literature not only does not substantiate his claims, but rejects them. See, *e.g.*, Nielsen *et al.*, 2013.). We need not discuss the details of this account further because insofar as they require that RECS implicate mechanistic explanations, they represent a misunderstanding of Chemero's account.

NONMECHANISTIC DYNAMICAL EXPLANATIONS

Chemero 2009 argues that many cognitive and perceptual phenomena can be explained without mental gymnastics, using embodied dynamical models in which the organism and its environment are viewed as coupled dynamical systems. It's a pretty humble claim, for all the controversy it generates; the only thing that the RECS advocate needs to demonstrate is that there are phenomena of interest to cognitive science that can be explained by nonmechanistic dynamical models. But, as we have seen, it is easy to allow the temptation to accept dynamical models of cognition *only insofar* as they can be given an accompanying mechanistic reading to operate as an influential, but unargued, assumption. There are two ways that we will approach defending RECS

against this requirement. The first will be a kind of burden-shifting. After all, there are existing dynamical explanations of perception and cognition that are not mechanistic (we'll look at one). We ask: if we already have a dynamical explanation of the cognitive phenomenon, why do we need the mechanistic explanation? Our second argument is both more interesting and gets more directly to heart of what makes RECS distinct from the mechanistic tradition; the claim that there are cognitive phenomena that are can be best viewed as systems that span nervous systems, bodies, and their environment. Instead of seeing the brain as a machine and the central locus of control, a view emerges in which the nervous system, body, and environment are engaged in reciprocal interplay of causal interactions in which no one element is best viewed as the "controller." In cases like this, we will argue, there are reciprocal exchanges of influence across the entire system – where the dynamics of the system's parts are simultaneously determining and being determined by the system's global dynamics. As we'll see, these systems resist mechanistic decomposition into the analogues of cogs and wheels. Moreover, what is equally troubling, dogmatic insistence on mechanistic approaches could (by drawing boundaries) act as a barrier to discovery for systems like this.

CASE 1: THE SUSSEX ROBOT

Our first case comes from robotics researchers Harvey, Husbands & Cliff (1994), who used an artificial evolutionary algorithm to generate a recurrent ANN architecture with a unique set of connections that were then evaluated for performance of a certain task as well as a particular visual morphology (the placement of its photoreceptors). The evolved visio-motor control systems were evaluated on the basis of their success at the specified task and the most successful among the population (according to their worst trial) would be given a higher chance of passing on their "chromosomes" to the next generation. The endgame of this selection process are robot control systems that are adept at the perceptual-motor task for which they were selected, but whose skillful engagement with the environment does not result from the researchers' constructing a preferred cognitive architecture as an optimal solution for coordination task. In fact, it is the other way around; rather than design a mechanism based on their conception of how an artificial agent might, for example, use simple visual stimuli to move itself toward a target, the roboticists allowed the agent's performance to home in on neural architectures whose dynamics reliably yielded successful action.

Before we look at the explanations available in specific instances, there are a couple of points worth noting. First, in their analysis of the robots' design, the authors do provide a brief description of the neural mechanisms and bodily structure of their evolved agent (Husbands, Harvey & Cliff, 1995). Still, these mechanistic-looking descriptions are restricted to invariant properties common to all of the evolved agents, such as the mechanism by which excitatory channels inhibit inhibitory activations and the angle of acceptance of the robot's photoreceptors. The analysis of these features explains, for example, the method by which their dynamical model of the control system was constructed by extending their mathematical treatment of feedback loops to the entire system; as we will show, *it's the dynamics of the control system, and its*

coupling with the egocentrically defined environment which are doing the explanatory work in understanding the robot's success. The second point to press is that the authors explicitly view robots “or any organism, as a physical dynamical system, with its own internal dynamics, coupled to a world (also a dynamical system) through sensory-inputs and motor-outputs (*ibid.*, p. 4)” and remain agnostic as to the utility of mechanistic/functional decomposition in such systems. With this in mind, let's look at the author's dynamical account of the robot's performance.

The Sussex roboticists' maintain that a full explanation of the control systems' achievement of general solutions to a task like navigation toward a target involves capturing the dynamical coupling that obtains between the “visual structure of the robot's world determined by its evolved visual morphology” and the internal dynamics of its neural control system (*ibid.*, 14). This is done by producing an egocentric mathematical characterization of the robots environment – values that define the robot's world in terms of the robot's position as well as its visual morphology. This was done by defining the robot's world (a circular arena) in terms of coordinates in what was dubbed “ $r\phi$ space,” with r being the robot's distance from the center of the arena and ϕ being the clockwise angle between the arena's center and the front of the robot. Given any particular visual morphology, $r\phi$ specifies every possible visual input of the robot in that environment. The second step in their dynamical account is to provide an accompanying mathematical characterization of the evolved network's dynamics. This was done first by examining the way that *generator units* – units that produce excitatory activity regardless of visual stimuli – contribute to feedback loops across visual input units, a number of motor units, as well as motor-output units that also form recurrent connections with visual input units. Extending this mathematical analysis to the entire network, the motor behavior of the robot can be determined for every possible visual input.

With these mathematical descriptions in hand, the Sussex roboticists then combined these two mathematical treatments into a single model that captured the coupled agent-environment system. Because every location in $r\phi$ space specified a visual input and every visual input (along with prior motor outputs, due to recurrence) yielded a determinate motor behavior, the roboticists were able to construct phase portraits that predict the behavior of the robot in any possible location and stage in the task. Such phase portraits also allow one to predict and understand the interplay between bodily structure and environmental structure as the robot moves through the arena. The activation of the robot's left and right motors is determined by its visual inputs and earlier motor outputs while the action of the motors, in moving the robot through its environment, is simultaneously determining the visual inputs it receives. Characterizing these dynamics in terms of locations in $r\phi$ space yields a phase portrait with a single attractor (the target location) and locates all possible coordinates in the phase portraits within the basin of that attractor, predicting the precise trajectory of the robot to its goal from any initial place in the arena. Moreover, the phase portrait predicts the patterns in behavior that explain the success of this particular evolved control system – for instance, the fact that the absence of a strong activation in either visual input will allow the endogenous excitatory activation of the generator unit to initiate a feedback loop that results

in circular rotation until a strong activation is found (appropriate to “looking around” for the target).

By our lights, as well as the Sussex roboticists', the phase portrait predicts and explains the behavior of the robot's skillful navigation of the task domain. It also explains why this neural architecture might have had a leg up in the evolutionary competition with its alternatives. Why, then, would we need a further mechanistic explanation? It is possible to provide a mechanistic account of how this particular robot successfully negotiates this perceptual task. One might, for instance, wish to explain why the robot rotates when “looking around,” citing various processes of excitation, inhibition, feedback etc. Chemero (2009) does just this:

Nodes 0 and 1 take input from separate visual fields (V1 and V2, respectively). When there is strong input in V1, whatever V2 is like, node 0 causes a pattern of activation that speeds up the left and right motors, causing motion straight ahead. When there is strong input to V2 and weak input to V1, node 1 causes a pattern of activation that leads to increased excitement in node 14, which excites itself and slows the left motor. This causes the robot to rotate in a circle, until a strong input is found in V1. (p. 74)

There is nothing obviously wrong with this mechanistic explanation of how the rotating behavior is generated as well as the contribution that such a mechanism would have in successful navigation toward the target – when the target is visible the robot would move forward and, when not, the robot would rotate until it comes into view. Still, it is not clear that any explanatory gains are made by augmenting the already existing dynamical account of the same phenomenon. After all, the same pattern of behavior is predicted in the thoroughly dynamical account. And, what's worse, the mechanistic account above was inspired by and relies heavily on the dynamical account already available, though the converse does not hold of the dynamical account. The dynamical perspective adopted by the Sussex roboticists quite accurately predicts the robots behavior under a wide range of conditions, explains the way the coupled system composed of agent and environment yields reliable coordination on the task, and reveals the dynamics of an evolved control system. What use is the mechanistic account when a mathematically precise model is already on offer?

CASE 2: HARRISON AND RICHARDSON'S PSEUDO-QUADRUPEL

Having attempted to shift the burden of proof, we will now make a positive case for accepting purely dynamical models as explanations of cognition, perception, and action. Our second case will motivate two main claims: 1) There are purely dynamical explanations that would be (and have been) considered within the purview of cognitive science when they were explained in terms of neural machinery and 2) Some dynamical models explain phenomena that seem to, by their very nature, resist mechanistic decomposition and would go undiscovered or underappreciated if a dogmatic insistence on mechanistic explanation were observed in cognitive science.

Steven Harrison & Michael Richardson's 2009 paper "Horsing Around" proposes that the synchronous oscillations commonly observed in locomotion is "the result of self-organization" and emerges from "nonlinear, yet lawful, multilevel interactions that occur among the many perceptual-motor components of biological movement systems" (Harrison & Richardson, 2009, p. 1). They investigated this by asking two subjects to walk on a treadmill together while causally coupled in various ways, and predicted that stable four-legged coordination patterns similar to those of natural quadrupeds would be produced spontaneously and that the stability of their gaits would depend on their degree of coupling. They also predicted that the speed of the treadmill might act as a control parameter for the entire person-person-treadmill system, determining when critical shifts in the overall dynamics of the system underwent qualitative changes. These predictions were made on the assumption that stable walking, jogging, running etc. could be understood as a special instance of coupled oscillator synchronization, and, accordingly, might be explained in the terms familiar to the general study of coupled oscillator systems. Their experiment put 12 naive participants into 6 pairs, both walking facing the same direction on a treadmill with one of the participants in the lead. The pairs were monitored in 4 coupling conditions. In the first of these conditions – the no coupling (NC) condition – the subjects were monitored walking by themselves at a comfortable pace. In the second condition, the rear subject was asked to maintain a distance of .75 meters behind his partner. This was known as the "visual coupling" or "VC" condition, since visual information was all the rear subject had at his disposal in maintaining the proper distance. The third condition was purely mechanical coupling (MC); the rear subject was blindfolded and attached to his partner by a .75 meter foam block. The final condition was a combined visual and mechanical coupling (VMC), that simply removed the blindfold from the rear subject – restoring the influence of visual information.

Each pair's walking patterns were given a measurement of *relative phase* (Φ) – the temporal relationship between their oscillatory cycles – determined by comparing one of the rear subject's leg movement with a target leg of her partner, a measure of their synchronization. Each limb's oscillation's "mean relative phase ($M\Phi$) and standard deviation of relative phase ($SD\Phi$) were used to assess the form and stability of the coordination (2)," with any two limbs being phase-locked when $SD\Phi < 20^\circ$. They found that a single collective variable could be used to portray the overall four-legged dynamical system: the relative phase of the back and front right limbs. Phase-locked walking patterns were categorized into one of four gaits, reflecting the total pattern of the four-legged dynamical system: pace (PACE), trot (TROT), lateral walk (LATW), or diagonal walk (DIAW), with each gait type corresponding to some range of the values for the order parameter – the value at which the system will undergo a transition from one stable pattern to a qualitatively different one – of the front and back right limbs ($M\Phi_{BR}$).

Interestingly, relative to the non-coupling condition, "VC, MC, and VMC couplings each [resulted] in an increase in the observed frequency of phase locked trials (*ibid.*, p. 3)." As was expected, an increase in the amount of coupling was correlated with a higher frequency of phase-locked locomotion,

with NC-pairings attaining phase-locked coordination (presumably by chance) in only 8% of trials, compared to 40% for VC, 63% for MC, and 77% in VMC. The observation of this significant increase in phase-locked four-legged suggests that coupling between the two participants (mechanical or visual) served as a basis for stabilizing the overall dynamics of the person-person-treadmill system. Furthermore, altering the speed of the treadmill decreased overall stability of the observed gaits; walking was phase-locked almost twice as often as jogging – in support of the view that speed could be treated as a control parameter for the system. Notably, PACE and TROT were observed almost exclusively and “the relative frequency of these patterns varied as a function of coupling (*ibid.*, p. 3).” That is, coupling resulted spontaneously in two characteristic patterns of synchronous oscillatory movement with different coupling conditions tending toward specific gaits. And, strangely enough, the coupled bipedal humans settled on patterns typical of biological quadrupeds.

Now comes the time to flesh out our two major (and related) claims. Harrison and Richardson's dynamical model of the pseudo-quadruped system is an exemplary case of a thoroughly dynamical explanation of a cognitive phenomenon – their model treated the person-person-treadmill system as a whole as a self-organizing dynamical system utilizing the same kinds of mathematical apparatus applied to spring-mass systems or pendulum swinging. If this does not seem to be a real case of cognition, bear in mind that the guidance of locomotion has certainly been *treated* as within the purview of cognitive and perceptual theorizing when it has been given a mechanistic explanation, as over a century of literature on central pattern generators (CPGs) in the nervous system attests (*e.g.* Brown, 1911; Guertin, 2009). If mechanistic explanations such as CPGs are explanations of cognitive phenomena, then purely dynamical alternatives like Harrison & Richardson's should be too, as they are of immediate relevance to the discussion. After all, if the pseudo-quadruped's rather unorthodox causal coupling is sufficient to establish and maintain stable quadrupedal locomotion, then the necessity of positing a specialized neural mechanism such as CPGs is greatly diminished. One might think that Harrison and Richardson's model is not an explanation at all, but merely presents a fascinating dynamical phenomenon crying out for a mechanistic explanation – this is the temptation mentioned earlier. But there are potential pitfalls here as well. As we argued in our last case, any mechanistic explanation that was tacked-on to such a purely dynamical account should prove itself in added explanatory power.

Unfortunately, though, there are further complications. Dynamical models like Harrison & Richardson's can be fiercely resistant to mechanistic decomposition, especially when they exhibit *interaction-dominant* dynamics. In interaction-dominant systems the dynamics of the individual components of the system determine the system's overall behavior, while these global features of organization oftentimes *simultaneously* condition the contribution and dynamics of the individual parts. The continuous feedback between the parts and the whole render the contribution of individual system components highly context-dependent – each part's activity both determines and is determined by the simultaneous activity of the other components, and the behavior of the components in the interaction-dominant system are quite different than in

isolation. This context-dependence makes it tough to assign a specific function to the individual parts of an interaction-dominant system – a difficulty that undermines attempts at mechanistic interpretations. If this is right, then insistence upon mechanistic explanations for the legitimation of dynamical approaches would establish a barrier to the discovery of interaction-dominant phenomena and would prejudge a number of nascent (but burgeoning) research programs. The point is, models that treat the coordination of locomotion as arising spontaneously in a multilevel interaction of bodily and environmental factors (and similar dynamical approaches) are not merely a mathematically flavored compliment to the mechanistic explanations on offer; sometimes they are the competition.

CONCLUSION

We conclude by returning to Ryle. In a section of *The Concept of Mind* (1949) called “The Bogy of Mechanism”, Ryle claims that “Men are not machines; not even ghost ridden machines.” (1949, p. 81). This points to the largest disagreement between Rockwell and RECS. Rockwell’s insistence on mechanistic explanation of cognition, and hence a version of representationalism, puts him in disagreement with Ryle and RECS. According to RECS and Ryle, cognitive agents are neither ghosts nor machines.

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